

Assignment-1

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Topic: Strings

Q.1A. Write a program that asks the user how many days are in a particular month, and what day of the week the month begins on (0 for Monday, 1 for Tuesday, etc), and then prints a calendar for that month. For example, here is the output for a 30-day month that begins on day 3 (Thursday):

M T W T F S S

1 2 3 4 5

6 7 8 9 10 11

Program:-

```
days=int(input('Enter the number of days in month: '))
weekDay=int(input('Day of the week the month begins on: '))

count=0
print('M\tT\tW\tT\tF\tS\tS')

for index in range(0,weekDay):
    print(end='\t')
    count+=1

for i in range(1,days+1):
    print(i, end='\t')
    count += 1
    if count%7==0:
        print(end='\n')
```

Q. 2. Check if all letters in a string are uppercase

Program:-

```
my_str=input('Enter the 1 line text: ')
count=0
my_str = my_str.replace(" ", "")

for i in my_str:
    if 'A'<=i<='Z':
        continue
    else:
        print('Not all letters in the string are uppercase')
        count+=1
        break
if count==0:
    print('all letters in the string are uppercase')
```

Q. 3. Write a version of a palindrome recognizer that also accepts phrase palindromes such as: Was it a rat I saw? or Dammit, I'm mad! Note that punctuation, capitalization, and spacing are usually ignored.

Program:-

```
my_str=input('Enter the phrase: ')
my_str=my_str.replace(" ", '')
my_str=my_str.lower()
str=''

for i in my_str:
    if 'a'<=i<='z':
        str=str+i

if str==str[::-1]:
    print('its palindrome')
else:
    print('its not palindrome')
```

Lists and Tuples

Q.1. Write a Python function that takes a list of words and returns the length of the longest one

Program:-

```
elements=int(input('Enter the number of words in the list: '))
list=[]
longLength=0
for i in range(0,elements):
    list.append(input(f'Word {i+1}: '))
    if len(list[i])>longLength:
        longLength=len(list[i])
print(list)
print('Length of the longest word in the list: ',longLength)
```

Q.2. Write a Python program to remove duplicates from a list

Program:-

```
elements=int(input('Enter the number of words in the list: '))
list=[]

for i in range(0,elements):
    list.append(input(f'Word {i+1}: '))
print('Before list:',list)
n=len(list)
i=0
while i<n:
    j=i+1
    while j<n:
        if (list[i] == list[j]):
            list.pop(j)
            n=len(list)
            continue
        j+=1
    i+=1
print('After list:',list)
```

Q 3. Create a list of books

e.g. booklist=[['Java 8', 700], ['Python for Beginners', 500],....]

Perform following operations on the list

1. Add a new book with price
2. Remove entry for a book
3. update price for a book
4. Sort the list by book names
5. Sort the list by prices
6. Print the book with max and min price [hint: you may use min()/max() functions of python]

Program:-

```
booklist = [['Java 8', 700], ['Python for Beginners', 500],
            ['JavaScript', 1000], ['C++', 600],
            ['C Programming', 550], ['.Net', 800]]
print('Books List: ',booklist)

#Add a new book with price
booklist.append(['HTML',400])
print('New book added: ',booklist)

#Remove entry for a book
booklist.remove(['C Programming', 550])
print('Book entry is removed from list: ',booklist)

#Update price for a book
booklist[1][1]=800
print('Updated price: ',booklist)

def get_price(book) :
    return book[1]

#Sort the list by book names
booklist.sort(key=str)
print('Sorted by book name: ',booklist)
```

```

#Sort the list by prices
booklist.sort(key=get_price)
print('Sorted by book price: ',booklist)

#Print the book with max and min price
max_book=max(booklist,key=get_price)
print('Maximum price: ',max_book)

min_book=min(booklist,key=get_price)
print('Minimum price: ',min_book)

```

Q.4. Write a Python program to compute element-wise sum of given tuples, using “zip()” function

Original tuples:

(1, 2, 3, 4)

(3, 5, 2, 1)

(2, 2, 3, 1)

Element-wise sum of the said tuples:

(6, 9, 8, 6)

Program:-

```

t1 =(1,2,3,4)
t2 =(3,5,2,1)
t3 =(2,2,3,1)

print(t1)
print(t2)
print(t3)

temp_tuple = []

for a, b, c in zip(t1, t2, t3):
    temp_tuple.append(a+b+c)

```

```
result = tuple(temp_tuple)
print('Sum of tuples: ',result)
```

Q.5 Write a Python program to find the repeated items of a tuple.

Program:-

```
t1=(1,2,3)
t2=(2,3,4)
temp_tuple=[]
print(t1)
print(t2)
for a in t1:
    for b in t2:
        if a==b:
            temp_tuple.append(a)

print(temp_tuple)
```

Outputs:-

String:-

Q1A.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day2\Assignment\Q1.py
Enter the number of days in month: 30
Day of the week the month begins on: 5
M  T  W  T  F  S  S
      1  2
3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30

Process finished with exit code 0
|
```

Q2.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day2\Assignment\Q2.py
Enter the 1 line text: Python is Dynamic Programming Language
Not all letters in the string are uppercase

Process finished with exit code 0
```

Q3.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day2\Assignment\Q3.py
Enter the phrase: Dammit, I'm mad!
its palindrome

Process finished with exit code 0
|
```

List and Tuples:-

Q1.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day3\Assignment\Q1.py
Enter the number of words in the list: 5
Word 1: Python
Word 2: Java
Word 3: JavaScript
Word 4: HTML
Word 5: C++ Programming
['Python', 'Java', 'JavaScript', 'HTML', 'C++ Programming']
Length of the longest word in the list: 15

Process finished with exit code 0
```

Q2.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day3\Assignment\Q2.py
Enter the number of words in the list: 7
Word 1: Python
Word 2: Java
Word 3: Python
Word 4: Python
Word 5: HTML
Word 6: C
Word 7: Java
Before list: ['Python', 'Java', 'Python', 'Python', 'HTML', 'C', 'Java']
After list: ['Python', 'Java', 'HTML', 'C']

Process finished with exit code 0
```

Q3.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day3\Assignment\Q3.py
Books List: [['Java 8', 700], ['Python for Beginners', 500], ['JavaScript', 1000], ['C++', 600], ['C Programming', 550], ['.Net', 800]]
New book added: [['Java 8', 700], ['Python for Beginners', 500], ['JavaScript', 1000], ['C++', 600], ['C Programming', 550], ['.Net', 800], ['HTML', 400]]
Book entry is removed from list: [['Java 8', 700], ['Python for Beginners', 500], ['JavaScript', 1000], ['C++', 600], ['.Net', 800], ['HTML', 400]]
Updated price: [['Java 8', 700], ['Python for Beginners', 800], ['JavaScript', 1000], ['C++', 600], ['.Net', 800], ['HTML', 400]]
Sorted by book name: [['.Net', 800], ['C++', 600], ['HTML', 400], ['Java 8', 700], ['JavaScript', 1000], ['Python for Beginners', 800]]
Sorted by book price: [['HTML', 400], ['C++', 600], ['Java 8', 700], ['.Net', 800], ['Python for Beginners', 800], ['JavaScript', 1000]]
Maximum price: ['JavaScript', 1000]
Minimum price: ['HTML', 400]

Process finished with exit code 0
|
```

Q4.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day3\Assignment\Q4.py
(1, 2, 3, 4)
(3, 5, 2, 1)
(2, 2, 3, 1)
Sum of tuples: (6, 9, 8, 6)

Process finished with exit code 0
|
```


Q5.

```
C:\Users\P6CP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\P6CP-AI\Python\Day3\Assignment\Q5.py
(1, 2, 3)
(2, 3, 4)
[2, 3]

Process finished with exit code 0
```